Run report: cen-q5-nonsplit-run-20261024T2210Z

Descriptive record of one recorded run of examples/genus4_n4_census.rs, part CENSUS (sampled) under `research/genus4_n4_census_20261024/PROTOCOL.md` (with its dated amendments). It lists what ran, on what host, and what the tool recorded, file by file. Interpretation is in the study's `REPORT.md`. Costs are counted operations (F_p multiplications and inversions, group operations, native operations), never wall-clock; the only times are the tool's own seconds. Isolation: none earned (a shared cloud container, or a RunPod CPU pod whose lscpu is in host.txt); this affects only the seconds, since every recorded value is an exact count or a counted operation total.

env.txt

```
run: cen-q5-nonsplit 20261024T2210Z
git HEAD: 7a579ce24983ef78f6af4ecf3dc9a8c6f10e8a91
tool: examples/genus4_n4_census.rs sha256 cc0bcbe7b428feff4e2acbea3c6887d6abb5031b01f3a49589bd3597b3773dc7
protocol: research/genus4_n4_census_20261024/PROTOCOL.md sha256 457dc80ceded9acbbb0fb39ca1c61e8c3e2bfa773f863100584969c76131187d
command: /tmp/claude-0/g4/g4 research/genus4_n4_census_20261024/results/cen-q5-nonsplit-run-20261024T2210Z CENSUS 5 nonsplit 50000000 202610245 0 2000
rustc: rustc 1.97.0 (2d8144b78 2026-07-07)
host: Linux 6.18.44-fc-v153 x86_64, 4 cores, Intel(R) Xeon(R) Processor @ 2.10GHz
threads: 1
wall seconds: 893
```

tool_log.txt

```
nonsplit 5: 6463588/50000000 samples, res-shaped 12302, allowed-trace cand 280, smooth hits 280, 120s
nonsplit 5: 13193329/50000000 samples, res-shaped 25149, allowed-trace cand 585, smooth hits 585, 240s
nonsplit 5: 19960616/50000000 samples, res-shaped 38235, allowed-trace cand 917, smooth hits 917, 360s
nonsplit 5: 26787134/50000000 samples, res-shaped 51318, allowed-trace cand 1252, smooth hits 1252, 480s
nonsplit 5: 33622218/50000000 samples, res-shaped 64347, allowed-trace cand 1539, smooth hits 1539, 600s
nonsplit 5: 40279645/50000000 samples, res-shaped 77305, allowed-trace cand 1854, smooth hits 1854, 720s
nonsplit 5: 47037410/50000000 samples, res-shaped 90330, allowed-trace cand 2158, smooth hits 2158, 840s
nonsplit q=5: samples=50000000 (random) res_shaped=96093 allowed_trace_candidates=2305 smooth_prime_order_hits=2305 smooth_fraction_among_prime_cands=1.0000 smooth_fraction_other_res_shaped(sub-
sample 2000)=1067/2000 secs=893.0
nonsplit q=5: total prime-order mass estimate 56.365 +- 1.174 (hit rate per sampled form 4.610e-5)
```

summary.txt

```
nonsplit q=5: samples=50000000 (random) res_shaped=96093 allowed_trace_candidates=2305 smooth_prime_order_hits=2305 smooth_fraction_among_prime_cands=1.0000 smooth_fraction_other_res_shaped(sub-
sample 2000)=1067/2000 secs=893.0
nonsplit q=5: total prime-order mass estimate 56.365 +- 1.174 (hit rate per sampled form 4.610e-5)
```

cand_t_hist.csv (first 12 of 84 rows)

type	t	candidates
nonsplit	−624	54
nonsplit	−49	119
nonsplit	−48	134
nonsplit	−46	541
nonsplit	−45	276
nonsplit	−44	435
nonsplit	−43	414
nonsplit	−42	270
nonsplit	−41	203
nonsplit	−40	1266
nonsplit	−39	1366
nonsplit	−38	385

hits.csv (first 12 of 2305 rows)

type	code	t	nE	smooth

nonsplit	31396477592	−27	653	true
nonsplit	126321324294	−27	653	true
nonsplit	133272213391	−33	659	true
nonsplit	101374332184	−27	653	true
nonsplit	142637028771	13	613	true
nonsplit	37515789705	−27	653	true
nonsplit	45304123164	−27	653	true
nonsplit	146657495888	33	593	true
nonsplit	94124475714	−27	653	true
nonsplit	109986340912	7	619	true
nonsplit	138799106113	−27	653	true
nonsplit	112206438948	7	619	true

mass_by_t.csv (first 12 of 12 rows)

type	t	nE	smooth_hits	mass_estimate	mass_stderr
nonsplit	−33	659	175	4.2793	0.3235
nonsplit	−27	653	753	18.4133	0.6710
nonsplit	−21	647	205	5.0129	0.3501
nonsplit	−17	643	149	3.6435	0.2985
nonsplit	7	619	212	5.1841	0.3560
nonsplit	9	617	47	1.1493	0.1676
nonsplit	13	613	221	5.4042	0.3635
nonsplit	19	607	36	0.8803	0.1467
nonsplit	25	601	353	8.6320	0.4594
nonsplit	27	599	34	0.8314	0.1426
nonsplit	33	593	81	1.9807	0.2201
nonsplit	39	587	39	0.9537	0.1527

Graph

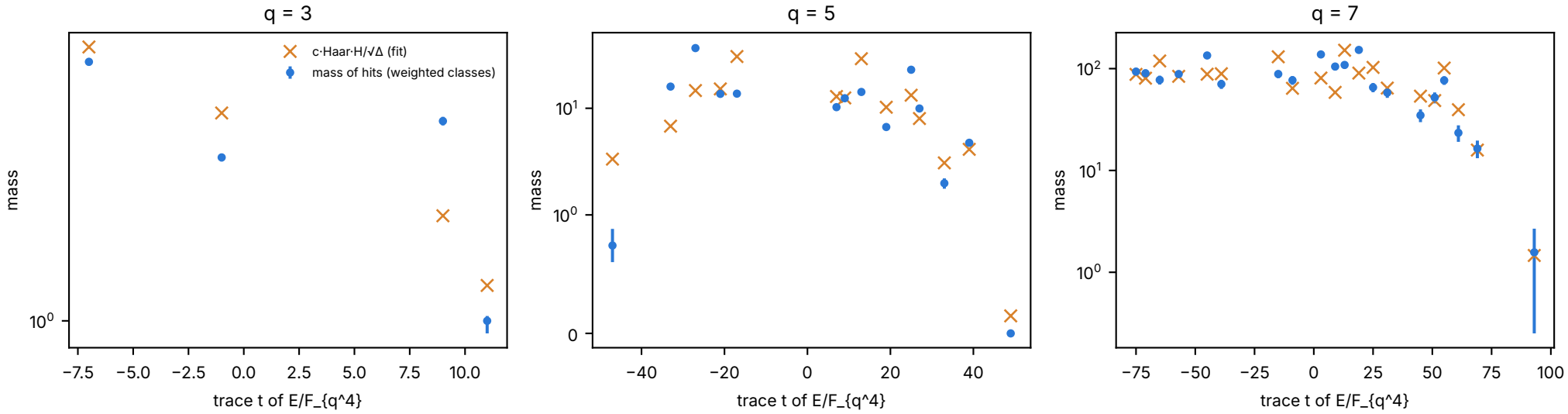


Figure 1: Mass of hits per prime-order trace with the Haar-density fit.